

Chik Kin Hei

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Education

Nanyang Technological University – Bachelor of Computing in Computer Science	Expected Jul 2030
St. Paul's College Hong Kong – 4 A* in Edexcel International A-Levels (Mathematics, Further Mathematics, Physics, Chemistry)	Jun 2026

Projects

Tacit Collusion in LLM Bidders | Python, pytest | Oct 2026 github.com/linusc9516/bid-pricing-collusion

- Tested whether LLM bidders tacitly collude by running 190 sessions across 3 models and 3 tie-break rules for about \$6, building a harness with a deterministic auctioneer and a matched one-shot control for every condition.
- Found no tacit rotation at 5 sessions per condition. Ties occurred in 3.07% of rounds against 0.03% by chance, and one two-bidder session held bids at 94–96 from round 10, so I flagged it for follow-up.
- Measured collusion with an index normalised to the Bayes-Nash benchmark (0 competitive, 1 at the reserve price), using the session as the unit and pre-specified Holm-corrected comparisons, so that overbidding is not mistaken for coordination.
- Added a pytest suite, log validation, resumable runs and a viewer that shows each session beside its control.

International Olympiad in Artificial Intelligence (IOAI) 2025 | PyTorch, NumPy, sklearn | Aug 2025

- Placed 2nd among all participants on a segmentation task by building a PyTorch model that handled a 50× class imbalance, raising accuracy on the critical human/wall classes by 37% through data augmentation and tuned CrossEntropyLoss weights.
- Solved 6 machine learning tasks (NLP, CV, deep learning) solo in a 2-day contest under 12-hour time and memory limits.
- Delivered a pick-and-place pipeline for the Galbot G1 robot with a team in a MuJoCo simulation, using A* path planning and inverse kinematics to navigate obstacles and organise objects on shelves.

Soular | React Native, TypeScript, Supabase | May 2025 github.com/endernoke/soular

- Cut average post load time by 46% for 100+ concurrent users by optimising Supabase queries and adding client-side data caching to the social feed and group chat.
- Cut Android build times by 65% by streamlining the CI/CD pipeline with Expo and Gradle optimisations, speeding developer feedback during hackathon sprints.
- Built a full-stack React Native mobile app with a team to help users discover environmental initiatives, with an AI assistant and a carbon footprint calculator for instant personalised insights.

Ulcerative Colitis Detector | PyTorch, ResNet, Flask, TorchScript | Jul 2025 github.com/linusc9516/UC_Classifier

- Reached 96% validation accuracy on an 11,000+ image colonoscopy dataset by fine-tuning ResNet18 with data augmentation, weighted random sampling and staged unfreezing.
- Built a Flask app for multi-image upload and real-time inference that aggregates predictions into a High/Medium/Low/Normal risk rating, simulating a clinical decision-support tool for gastroenterology.
- Sped up CPU inference by tracing the model with TorchScript, so it deploys without the PyTorch training environment.

Honors & Awards

Bronze Medal, International Olympiad in Artificial Intelligence (IOAI) – top score of 99.62 on one task	Aug 2025
2nd Place and Innovation Award, JA Code for Impact – Soular	May 2025
Gold Award, IOAI 2025 Hong Kong Qualifier Competition	2025

Skills

Languages: Python, TypeScript, JavaScript, C++, C, SQL

ML & Research: PyTorch, TransformerLens, inspect_ai, JAX, NumPy, Pandas, scikit-learn, Matplotlib

Tools & Platforms: Git, uv, pytest, Flask, React Native, Supabase, Expo

Certificates

CS50x: Introduction to Computer Science – HarvardX Aug 2024

Microsoft Certified: Azure AI Fundamentals – Microsoft Jul 2021